Design, Implementation and Analysis of 4-Bit Look Ahead Carry Adder using Adiabatic and Stacking Techniques

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***Abstract*—For many years, designing of high speed low power circuits with CMOS technology was a difficult challenge for research community. There are various levels at which design problem related to low power and increased demand can be addressed; these levels are - software level, architecture level, algorithm level, circuit level and process technology level. In this paper we have designed a 4-bit Look Ahead Carry Adder using CMOS technology. The objective of this paper is to put different approaches that reduce the consumption of power of Look Ahead Carry Adder. Conventional Look Ahead Carry Adder is compared with Adiabatic and Stacking Look Ahead Carry Adder.**

Keywords—CMOS Circuit, VLSI, Combinational Circuits, Tanner EDA, Power.

I. INTRODUCTION

Industry over last few decades has seen a tremendous growth in the electronics segments. Market is flooded with highly sophisticated and technically advanced gadgets [7]. The advances in electronics industry because of technically advanced ICs (Integrated Circuits) and increase in transistor switching speed has resulted in many fold performance improvement in computer systems. On the hind side, there has been increased consumption of power to bring this change. A community of scientists / researchers has thus being trying to reduce this dissipation by improving the circuit design [12], [13].

High performance systems cause high energy consumption, need better cooling technology, require expensive packaging, decrease system reliability and on top of all-increase the cost of system.

Power depends on the way circuit is designed however it can be divided into static and dynamic power. The former is generated due to direct current bias, as in case TTL (Transistor–transistor–logic) and ECL (emitter–coupled–logic) and NMOS (N-type MOS) or due to leakage current. Except for push – pull type logic family such as CMOS, static power tends to dominate [12], [13], [14]. Because of this CMOS is considered to be most suitable circuit style for VLSI (very large scale integrated circuit) [10].

The driving force behind Very Large Scale Integration circuits (VLSI) for high performance computing related to science and technology has been digital CMOS integrated circuits. It is speculated that the demand for digital CMOS IC's will keep increasing because of its important feature improvement in processing technology, reliable performance and low power [14].

ALU is the basic building block of the CPU and Adder is the very important part of the ALU. It is used for many functions like addition, multiplication etc [4]. Adders are used not only in the ALU, but also in other parts of the processor, where they used to calculate addresses and some other part of the processor. In ripple carry adder, the delay time is major concern. Reducing the carry propagation delay of adder is the very important [3]. Different logic designed approaches have been employed to overcome the carry propagation problem. One widely used approach the principle of Look Ahead Carry Adder solves this problem by calculating the carry signal in advanced, based on input signal [4]. Look Ahead Carry Adder is mostly used in the digital systems. It improves the speed and reduces the power consumption. In this paper Carry Look Ahead Adder is designed with several techniques (Adiabatic and Stacking).

II. LOOK AHEAD CARRY ADDER

The Look Ahead Carry adder is based on the concept of propagation and generating the carry bit. It calculates the one or more carry bits before the sum, which reduces the wait time to calculate the result of larger value bits [3], [4], [6].

It is based on the fact that a carry signal will be generated in two cases [4].

1. When both A_i and B_i are 1.
2. When one of the two bits is 1 and the carry-in (carry of the previous stage) is 1.

To understand the Carry propagation problem, let's consider the case of adding two n-bit numbers A and B.

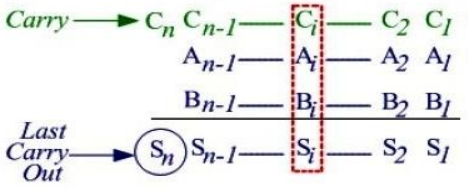


Fig. 1 Addition of two n-bit numbers

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Full adder circuit used to add the operand bits in the i^{TH} column; namely A_i & B_i and the carry bit coming from the previous column (C_i) [11].

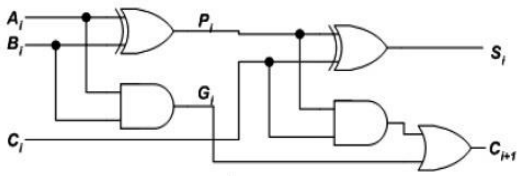


Fig. 2 Full Adder circuit diagram

In this circuit 2 internal signal P_i and G_i are given as [1], [2]:

$$P_i = A_i \oplus B_i$$
$$G_i = A_i B_i$$

The output sum and carry can be defined as [1], [2]:

$$S_i = P_i \oplus C_i$$
$$C_{i+1} = G_i + P_i C_i$$

G_i is known as the carry generate signal, since a carry C_{i+1} is generated whenever G_i=1, regardless of the input carry C_i [2].

P_i is known as the carry propagate signal, since whenever P_i=1, the input carry is propagated to the output carry, i.e. $C_{i+1} = C_i$ (whenever $P_i = 1$ & $G_i = 0$). Computing the values of P_i and G_i only depend on the input operand bits (A_i & B_i). Thus, these signals settle to their steady-state value after the propagation through their respective gates. Computed values of all the P_i's are valid one XOR-gate delay after the operands A and B are made valid. Computed values of all the G_i's are valid one AND-gate delay after the operands A and B are made valid [2], [11].

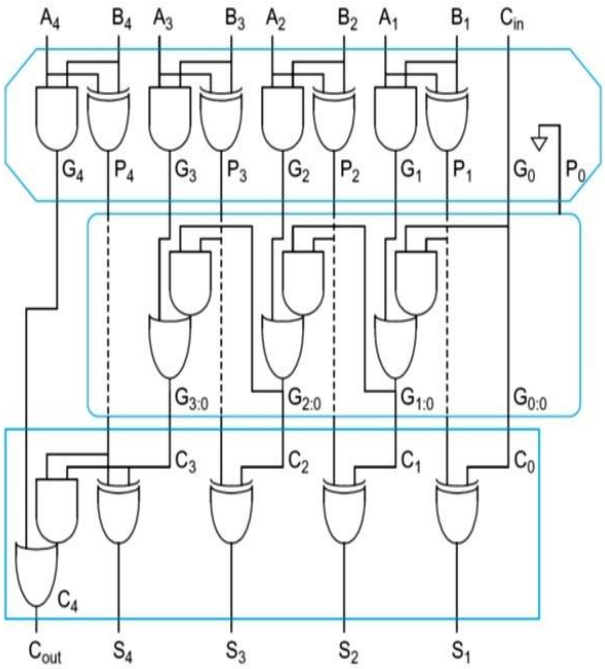


Fig. 3 Look Ahead Carry Adder Circuit Diagram

The Boolean expression of the carry outputs of various stages can be written as follows [2], [5]:

$$C_1 = G_1 + P_1 C_0$$
$$C_2 = G_2 + P_2 C_1 = G_2 + P_2 (G_1 + P_1 C_0)$$
$$= G_2 + P_2 G_1 + P_2 P_1 C_0$$
$$C_3 = G_3 + P_3 C_2 = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 C_0$$
$$C_4 = G_4 + P_4 C_3$$
$$= G_3 + P_4 G_3 + P_4 P_3 G_2 + P_4 P_3 P_2 G_1 + P_4 P_3 P_2 P_1 C_0$$

III. POWER REDUCTION DUE TO STACKING TECHNIQUE

Stack approach is one way of reducing power leakage which forces a stack effect by breaking down an existing transistor into two half size transistors. Structure is shown in fig.4. Reverse bias is induced between two transistors when they are turned off together resulting in sub threshold leakage current reduction. Delay is increase significantly by dividing transistors which could limit usefulness of the method [7], [8].

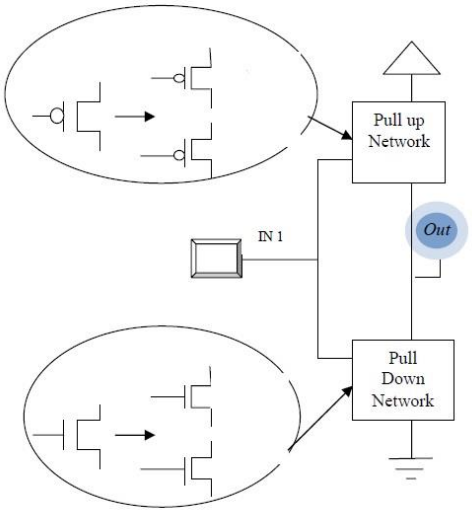


Fig.4 CMOS circuit using stacking approach

IV. POWER REDUCTION DUE TO ADIABATIC TECHNIQUE

Adiabatic is a term which is often used in thermodynamics to describe a process where there is no exchange of heat between environment and surrounding. Similarly adiabatic technique for reducing power implies no loss of heat in the circuit. The signal energies stored in the circuit capacitances are recycled instead of being dissipated as heat. The adiabatic logic is also known as energy recovery CMOS [6], [9].

There is a conservation of circuit energy with adiabatic switching approach. Hence this can be useful method to reduce power dissipation of digital system.

Adiabatic switching principle is one of the best Power reduction techniques having many advantages over other techniques like simple to design, no complexity increases, and great reduction in power dissipation [9], [14].

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Basically adiabatic switching principle states that during the transition from logic 0 to logic 1 we need the power supply but transition from logic 1 to logic 0 we don't need the power supply. If at that time we make the power supply off we can save the power dissipation because power dissipation is directly proportional to the supply voltage [9].

This can be done by changing the voltage source either by sinusoidal voltage source or by Pulse and adjusting the supply voltage in such a way that during the transition from logic 1 to logic 0 the supply voltage comes into off state and in this way the power dissipation is reduced up to much extent because as told earlier the power dissipation is directly proportional to supply voltage [6], [9].

V. SIMULATION AND WAVEFORM

4-Bit Look Ahead Carry Adder simulated circuits using conventional, adiabatic and stacking techniques.

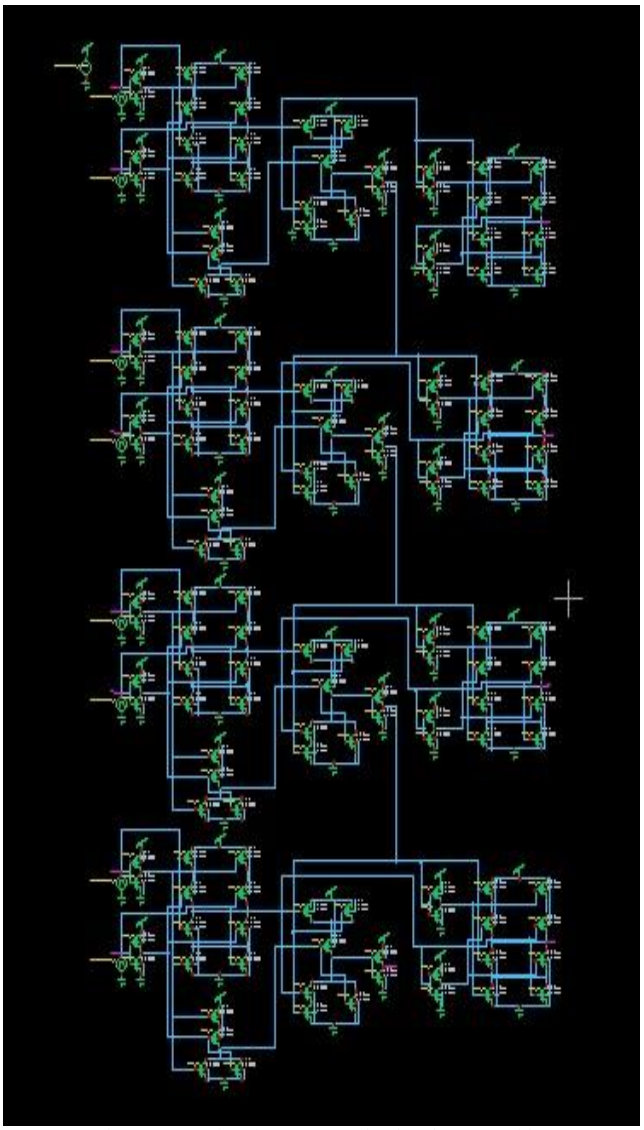


Fig.5 Conventional CMOS Look Ahead Carry Adder

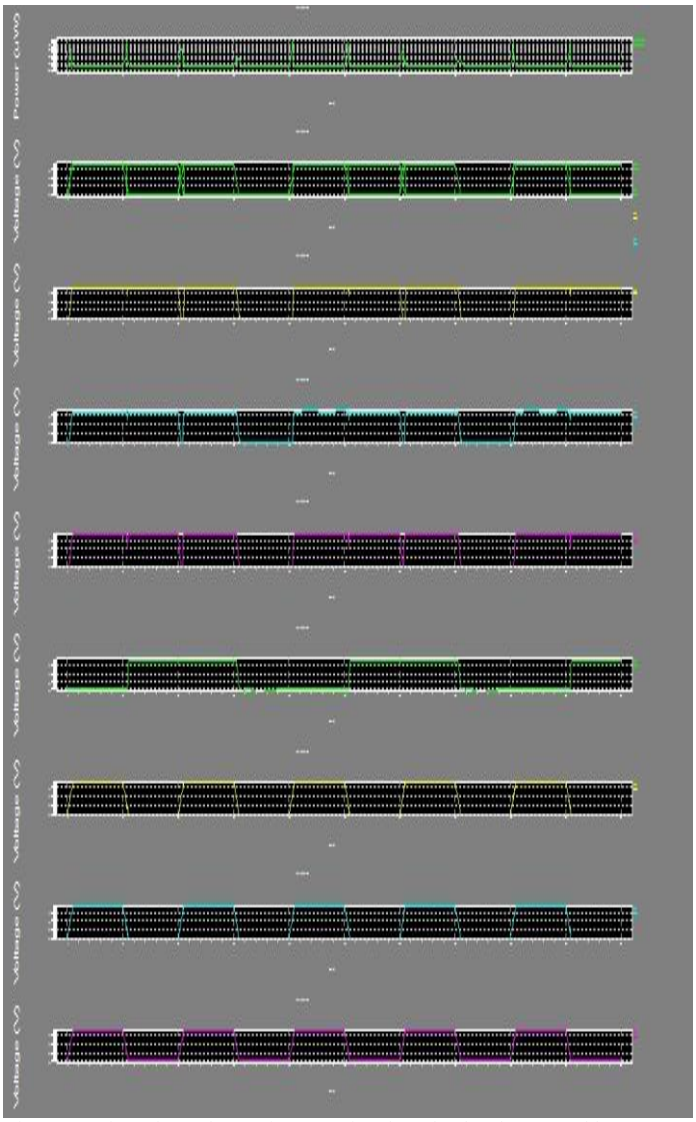


Fig.6 Waveform for Voltage of Conventional Look Ahead Carry Adder

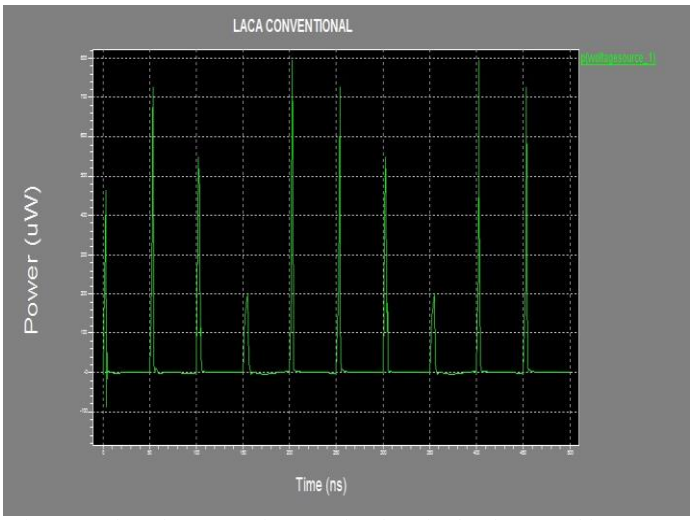


Fig.7 Waveform for Power of Conventional Look Ahead Carry Adder

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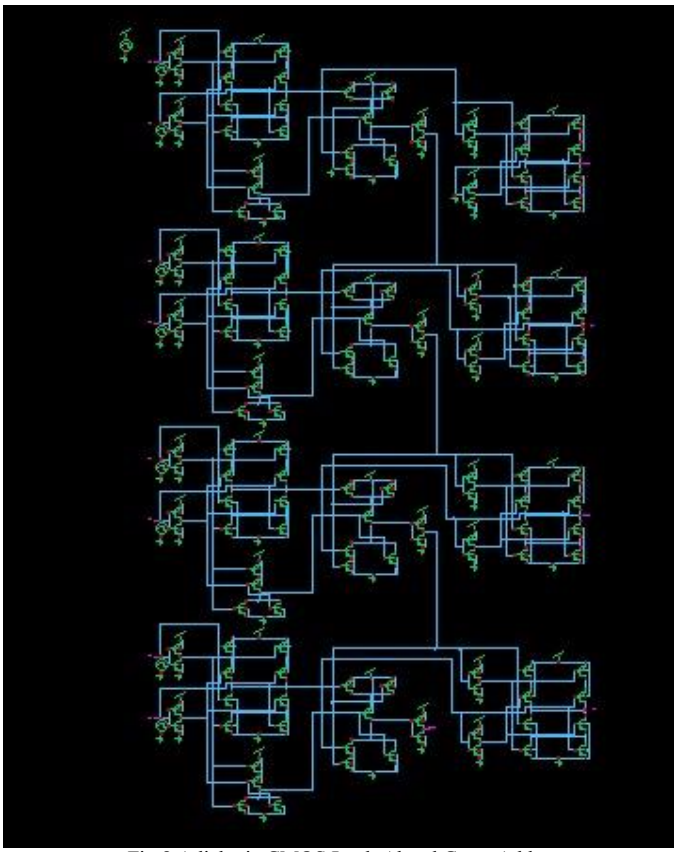


Fig.8 Adiabatic CMOS Look Ahead Carry Adder

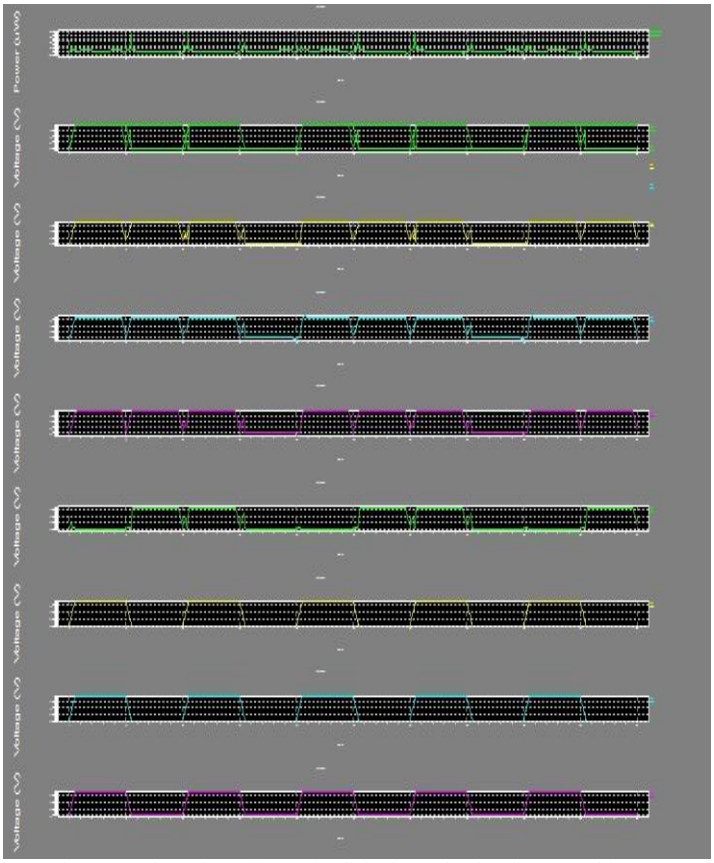


Fig.9 Waveform for Voltage of Adiabatic Look Ahead Carry Adder

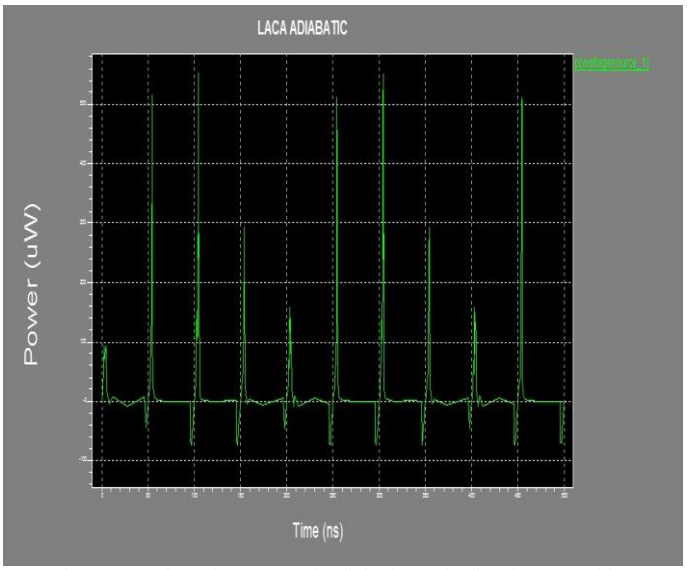


Fig.10 Waveform for Power of Adiabatic Look Ahead Carry Adder

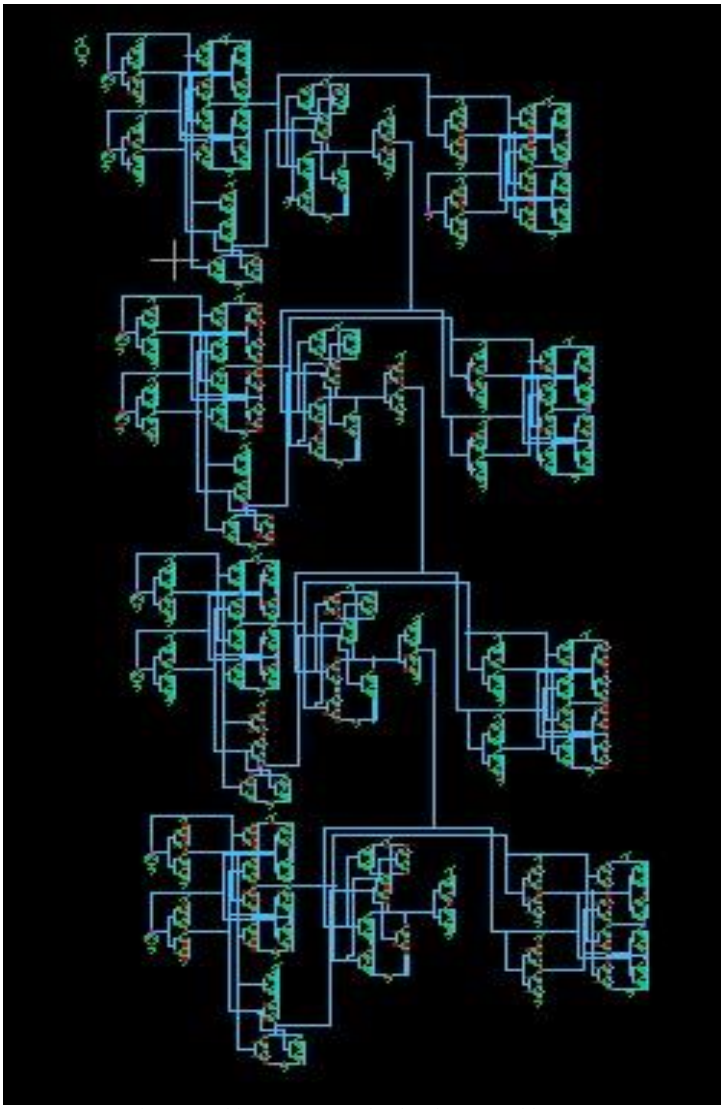


Fig.11 Stacking CMOS Look Ahead Carry Adder

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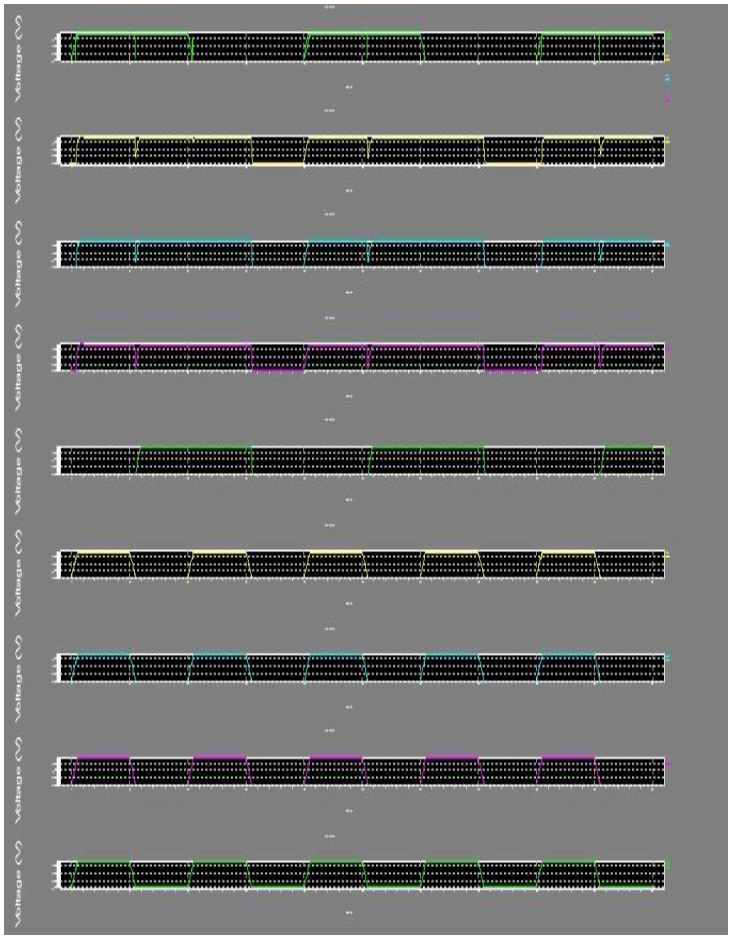


Fig.12 Waveform for Voltage of Stacking Look Ahead Carry Adder

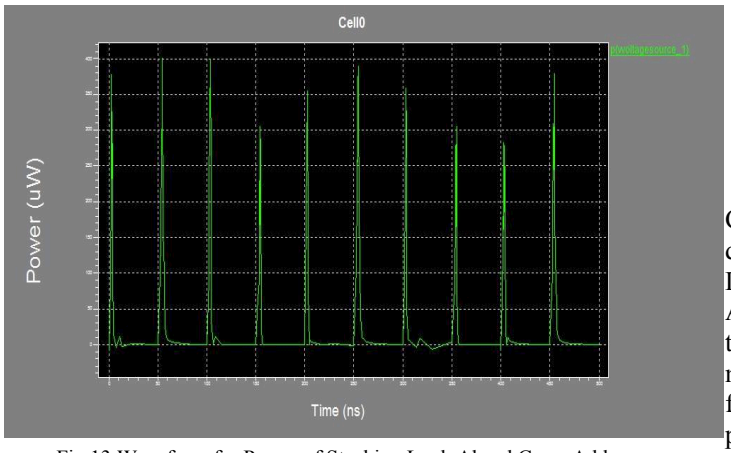


Fig.13 Waveform for Power of Stacking Look Ahead Carry Adder

VI. RESULT ANALYSIS

The Simulation of Look Ahead Carry Adder with conventional, adiabatic & stacking low power techniques are carried out at 180nm, technology. CMOS technology parameters are taken for NMOS and PMOS transistors, using Tanner tool. Transient Analysis is done to get Delay and Average Power results.

TABLE I. CMOS LOOK AHEAD CARRY ADDER RESULT

Power Reduction Technique	Average Power Consumption (In Microwatt)	Average Delay (In Nano Second)
Conventional	22.341	24.362
Adiabatic	7.705	27.176
Stacking	16.304	23.447

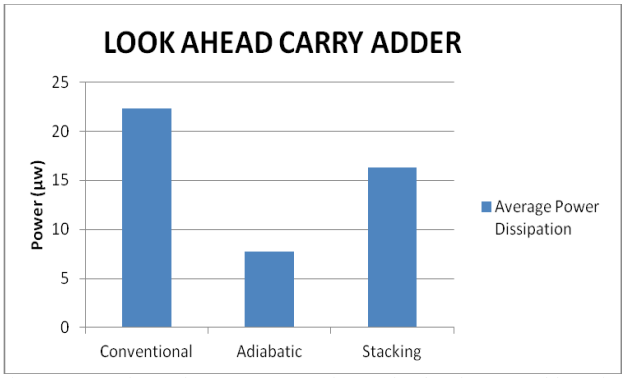


Fig.14 Comparison of Average power for Look Ahead Carry Adder using different Techniques

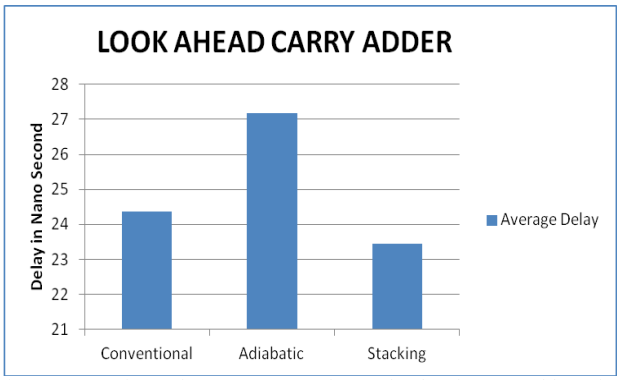


Fig.15 Comparison of Average power for Look Ahead Carry Adder using different Techniques

VII. CONCLUSION

Comparison has been done for Look Ahead Carry Adder, delay and power is calculated for Different Techniques. Designed the Look Ahead Carry Adder using Conventional, Adiabatic & Stacking Techniques and implemented these techniques for power reduction then we can observe the minimum power consumed in Adiabatic Technique. The tool for simulation is TANNER. At 180 nm technology and the practical observations have been tabled.

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